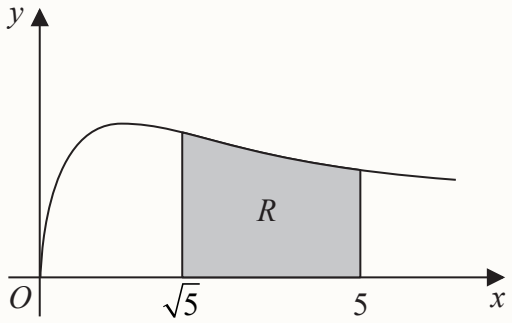



**Figure 1**

**In this question you must show all stages of your working.**

**Solutions relying entirely on calculator technology are not acceptable.**

Figure 1 shows a sketch of the curve with equation

$$y = \sqrt{\frac{3x}{3x^2 + 5}} \qquad x \geqslant 0$$

The finite region $R$, shown shaded in Figure 1, is bounded by the curve, the $x$-axis and the lines with equations $x = \sqrt{5}$ and $x = 5$

The region $R$ is rotated through 360° about the $x$-axis.

Use integration to find the exact volume of the solid generated. Give your answer in the form $a \ln b$, where $a$ is an irrational number and $b$ is a prime number.

**(5)**